weeks

2) Wireless Sensor Network Energy Optimization

Application: Wireless sensor network

Problem Statement: A wireless sensor network contains multiple sensor nodes with limited battery energy. Develop a ~~CS based Optimization algo~~ Grey Wolf optimization Algorithm to identify suitable communication parameters (or) routing configurations that reduce energy consumption while maintaining reliable data transmission.

Objective: Minimize network energy consumption and extend network lifetime while maintaining an acceptable packet delivary rate.

Psudocode

Begin

1. Initialize Parameters
   Set number of wolves (N), maximum iteration and Communication parameters.
   initalize Parameter a=2.
2. Initalize population.
   Generate random wolf positions representing different routing configurations and transmission power levels.
3. Evaluate fitness.
   Calculate the energy Consumption of each wolf.
   Evaluate the packet delivery rate.
   Identify the three best wolfs.

Alpha(α), Beta(β) and Delta(δ)

4. Update wolf Positions
   Update wolf positions based on Alpha, β and δ positions
   Calculate the new position using the GWO position update equations.

5. Iterate
   Decrease parameter a from 2 to 0.
   Repeat fitness evaluation and position updating until maximum iterations. (or) convergence is reached

6. Output Best Solution
   Display the best routing configuration
   Display minimum energy consumption and packet delivary rate.

End.

$x(t)$ = Current position of wolf being updated.

Distance vectors $(D_\alpha, D_\beta, D_\delta)$

$$\vec{D}_\alpha = |\vec{C}_1 \cdot \vec{X}_\alpha - \vec{X}(t)|$$
$$\vec{D}_\beta = |\vec{C}_2 \cdot \vec{X}_\beta - \vec{X}(t)|$$
$$\vec{D}_\delta = |\vec{C}_3 \cdot \vec{X}_\delta - \vec{X}(t)|$$

Potential next position (X_1, X_2, X_3)

$$\vec{X}_1 = \vec{X}_\alpha - \vec{A}_1 \cdot \vec{D}_\alpha$$
$$\vec{X}_2 = \vec{X}_\beta - \vec{A}_2 \cdot \vec{D}_\beta$$
$$\vec{X}_3 = \vec{X}_\delta - \vec{A}_3 \cdot \vec{D}_\delta$$

final update position

$$\vec{X}(t+1) = \frac{\vec{X}_1 + \vec{X}_2 + \vec{X}_3}{3}$$

output:

WIRELESS SENSOR NETWORK - GJWO

Enter number of wolves : 20
Enter Maximum iterations : 100
Enter minimum transmission power : 0.1
Enter Maximum transmission power : 1.0
Enter Minimum Communication distance : 2
Enter Maximum Communication distance : 10
Enter required PDR (in percentage : 90

---- optimization Result ----
optimal Transmission Power : 0.371
optimal Communication Distance : 10.0
Minimum Energy Consumption : 4.212
Packet Delivery Rate : 84.37%
Network Lifetime (relative) : 23.74
Status : Reliable Data Transmission.

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